International Trade[1]

Lecture Note 1: Gains of Trade in Neoclassical Models

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[1]The notes are still preliminary. Please, if you find any typo or mistake, send it to malfaro@ualberta.ca.

Contents

Notation

This is a derivation

This is some comment

This is a comment on advanced topics that are not part of the course (you can ignore it without loss of continuity regarding the text)

- The symbol ":=" means "by definition".
- I denote vectors by bold lowercase letters (for instance, $\mathbf{x}$) and matrices by bold capital letters (for instance, $\mathbf{X}$).
- To differentiate between the *verb* "maximize" and the *operator* "maximum", I denote the former with "max" and the latter with "sup" (i.e., supremum). The same caveat applies to "minimize" and "minimum", where I use "min" and "inf" respectively, with the latter indicating infimum.
- "iff" means "if and only if"
- $\exp(x)$ is equivalent to e^x.
- Random variables are denoted with a bar below. For instance, $\underline{x}$.

The notes include hyperlinks that take you directly to specific parts of the document. These links are displayed in both blue and red text. If you are using Adobe Acrobat Reader, you can easily return to your previous location by pressing Alt+Left Arrow.

1 Introduction

The first set of models that we will study in the course belong to the so-called **Neoclassical models**. They are among the the earliest attempts to formally explain why countries engage in international trade. These models share a common motive to trade: *differences between countries*, with a particular focus on differences in each country's supply side. For instance, all variants of Ricardian models consider differences in production technologies, while the Heckscher-Ohlin model emphasizes differences in factor endowments.

The goal of these notes is to establish a common result of Neoclassical models: all countries benefit from trade, regardless of the model's feature in which countries differ. In other words, Neoclassical models are unable to account for scenarios where countries are worse off under trade.

2 Neoclassical Models

The two key assumptions of Neoclassical models to entail positive gains of trade are perfect competition and differences between countries. Note that this does not mean that other model aspects are irrelevant. Rather, it means that such features are only significant for other predictions (e.g., patterns of production and trade), but not to predict increases in welfare due to trade.

To understand why perfect competition is crucial, recall that **goods are homogeneous** in this market structure. This implies that all goods possess identical observable and unobservable features. Consequently, **differences in prices are the sole aspect affecting a consumer's decision,** which in turn are entirely determined by marginal costs under perfect competition.

In this context, differences between countries in the supply side are essential, as they put some structure to the relative marginal costs across countries. Specifically, they lead to variations in the relative opportunity costs of producing goods across countries, and hence to the existence of Comparative Advantages.

Dixit and Norman (1980) provide a general definition of Comparative Advantage that applies to all Neoclassical models, regardless of the specific source of differences between countries. According to their definition, a country has a comparative advantage in producing a good if its autarky price (the price in a closed economy) is lower than that of another country. Notably, countries necessarily have comparative advantages in different goods, ensuring that each country can produce at least one good at a lower cost than others. This creates the foundation for mutually beneficial trade.

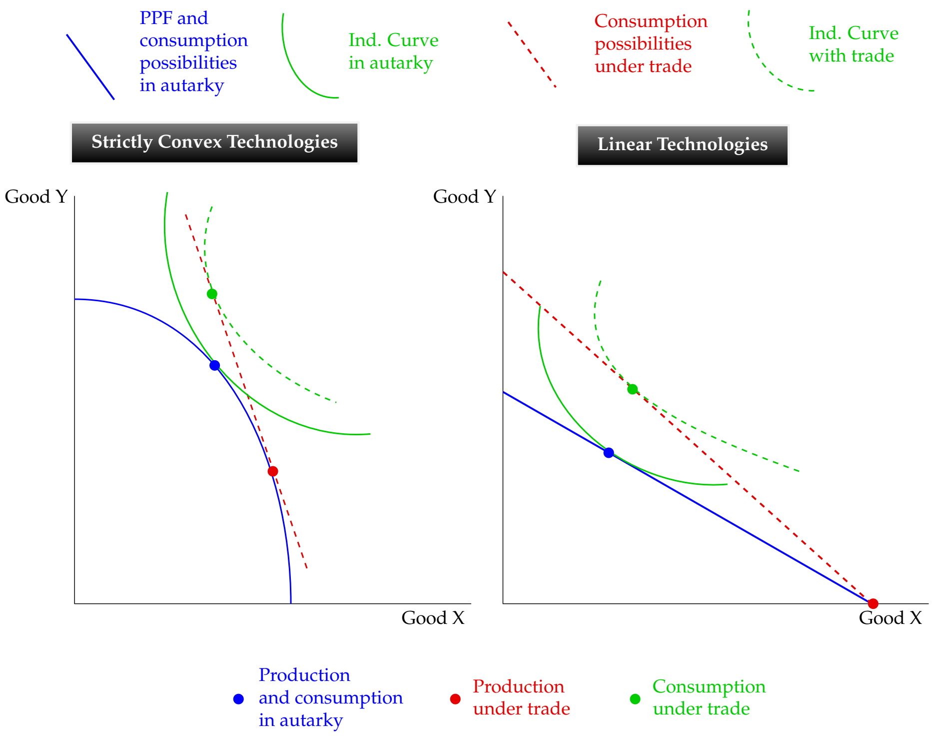
3 Gains From Trade in Neoclassical Models

The proof of gains from trade that we'll present relies on two further assumptions. First, demand allows for a representative consumer with non-satiated preferences. This assumption simplifies the analysis by aggregating the preferences of many individual consumers into one representative agent who always prefers more of a good to less.

Second, the model assumes balanced trade in equilibrium, meaning that the value of a country's exports equals the value of its imports. This in particular precludes the possibility of a trade deficit, which is equivalent to a country financing a foreign country's consumption. Notice that a trade surplus can be already ruled out as an equilibrium outcome, given that consumers are non-satiated and hence always seek to consume as much as possible.

The argument for gains of trade in each country is based on Pareto optimality. According to the Fundamental Theorems of Welfare, any allocation under perfect competition is Pareto efficient. Moreover, this allocation maximizes an agent's utility under balanced trade and subject to the economy's production constraints (i.e., resources and technology). Establishing gains of trade requires comparing a closed economy (autarky) with the same economy under trade (either totally or partially open). The proof follows **since autarky can be understood as a specific consumption allocation satisfying trade balance, where exports and imports are zero**. In other words, since autarky is feasible under trade, the optimal allocation under trade cannot make an agent worse by a revealed-preference argument.

Intuitively, trade expands a country's consumption choices following trade liberalization. Graphically, this can be observed by the increase in the possibilities of consumption under trade (red dashed line) relative to the consumption possibilities in autarky (blue solid line).



3.1 Source of Gains from Trade

Neoclassical models highlight the role of imports in a country as a means to increase welfare. In this context, exports have a specific interpretation within the model. This view is perfectly summarized by the following two quotes from **Milton Friedman**.

> "*Exports are the cost of trade, imports the return from trade, not the other way around.*"[1]

> "*Let's suppose that everything is cheaper in Japan. The Japanese sellers would be paid for them in dollars. What would they do with the dollars? Nothing for them to buy in the United States. If they would be willing to burn them up or to bury them in the Pacific Ocean, ah, that would be wonderful. After all, there is no product we can produce more cheaply than green pieces of paper.*"[2]

[1] Quote extracted from this article.

[2] Quote from "*Free Trade: Producer vs. Consumer*", lecture delivered at the Alfred M. Landon Lecture at Kansas State University 1978 April 27. The video is available here.

3.2 Formal Proof of Gains of Trade Using Duality (OPTIONAL)

In the following, we provide a formal proof for the existence of gains of trade, based on consumer duality. With this goal, let's keep assuming the existence of one representative agent. Moreover, using superscripts NT and T to denote "not-trading" and "trading", define:

- $\mathbf{q} := (q_1, q_2, ..., q_m)$ as the vector of net outputs
- $\mathbf{p} := (p_1, p_2, ..., p_n)$ as the vector of goods prices
- $\mathbf{c} := (c_1, c_2, ..., c_n)$ as the vector of quantities consumed
- $\mathbf{v}$ as the vector of inputs
- y as the consumer's income
- $e(\mathbf{p}, u)$ as the expenditure function to achieve utility u
- $\mathbf{p}^{NT}$ and $\mathbf{p}^{T}$ as the vector of prices under autarky and trade, respectively.
- $\mathbf{c}^{NT}$ and $\mathbf{c}^{T}$ as the consumption allocations under autarky and trade, respectively.

The proof makes use of an agent's minimum-expenditure function and the following fact:

$$\begin{aligned} e\left(\mathbf{p}^{T}, u^{NT}\right) &\leq \mathbf{p}^{T} \cdot \mathbf{c}^{NT} \\ &= \mathbf{p}^{T} \cdot \mathbf{q}^{NT} \\ &\leq r\left(\mathbf{p}^{T}, \mathbf{v}\right) \\ &= e\left(\mathbf{p}^{T}, u^{T}\right), \end{aligned}$$

which implies that $e\left(\mathbf{p}^{T}, u^{NT}\right) \leq e\left(\mathbf{p}^{T}, u^{T}\right)$. Using this and that e is increasing in the level of utility, $u^{T} \geq u^{NT}$ follows.

To explain the result, let's inspect the role of each line. By definition, the minimum-expenditure function is the value of the least expensive bundle that provides utility u^{NT}. Since $\mathbf{c}^{NT}$ allows the consumer to achieve the utility u^{NT}, the first line stipulates that the expenditure $\mathbf{p}^{T} \cdot \mathbf{c}^{NT}$ is either a solution (in which case the first line holds with equality) or a feasible bundle (since it provides utility u^{NT}). Thus, $\mathbf{p}^{T} \cdot \mathbf{c}^{NT}$ has to be strictly lower by definition of a solution, implying that consumers can derive the same utility as in autarky more economically.

The second line reflects that demand equals supply in equilibrium, so that $\mathbf{q}^{NT} =$

$\mathbf{c}^{NT}$. As for the third line, quantities maximize revenues in perfect competition. Thus, $r\left(\mathbf{p}^{T},\mathbf{v}\right)$ is higher than any other feasible revenue, in particular of that arising when $\mathbf{q}^{NT}$ is produced. The consequence is that the output value under trade is greater than when autarky quantities are produced.

Finally, the last line establishes that revenue under trade equals the country's expenditure, since trade is balanced. Thus, the maximum revenue under trade equals the minimum expenditure to get u^{T}.